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MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP (HONOURS)
Regular EXAMINATION October 2025
Third SEMESTER
Value Addition Course (VAC) - MG3VACFTQ200 - INTRODUCTION TO GOOD LABORATORY PRACTICES

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap) Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions Answer any 5 questions Each question carries 2 marks

1. Explain how the GLP guidelines revised over time? [U] / [CO1]

Good Laboratory Practice (GLP) refers to a set of principles intended to ensure the quality, reliability, and integrity of non-clinical laboratory studies. These guidelines provide a framework for the conduct, monitoring, recording, reporting, and archiving of laboratory studies, especially those related to safety testing of chemicals, pharmaceuticals, and food products. Overtime the GLP guidelines have evolved from basic laboratory conduct rules to comprehensive international standards ensuring data integrity, reliability, and regulatory compliance.

2. Why should chemicals never be pipetted by mouth? [C] / [CO5]

Mouth pipetting not only jeopardized the health of the scientists but also compromised the reliability of the results, as it was challenging to transfer precise volumes using this method. Moreover, there was a risk of contaminating the sample or reagent being pipetted with respiratory aerosols from the user

3. Write the formula for calculating normality [K] / [CO5]

Normality (N) = Number of gram equivalents / Volume of solution (in litres) Number of gram equivalents = Mass of solute (g) / Equivalent weight (g/eq)

4. Why must you use only appropriate containers when storing acids? [K] / [CO4]

Use only appropriate containers when storing acids because acids are highly reactive and corrosive, and the wrong container can lead to dangerous chemical reactions, leaks, or even explosions.

5. Describe the scope of GLP? [U] / [CO1] The scope of GLP (Good Laboratory Practice**)** refers to the range of activities, studies, and facilities to which GLP principles apply. GLP provides a quality framework for the organizational

process, personnel, facilities, equipment, and documentation involved in non-clinical laboratory studies that are intended to support research or product safety assessments.

6. Name the standard solution used for pipette calibration. [K] / [CO5]

The standard solution used for pipette calibration is typically distilled or deionized water. Water is used because its density and physical properties are well known and can be accurately corrected for temperature.

7. Write a short note on mathematical tools used in GLP? [A] / [CO2]

In Good Laboratory Practice (GLP), mathematical tools play a vital role in ensuring the accuracy, reliability, and validity of experimental data. These tools help in data analysis, error checking, and result interpretation — all essential for maintaining scientific integrity and regulatory compliance. **[2x5 = 10]**

Part B

Short Essay Questions Answer any 5 questions Each question carries 4 marks

8. Discuss why is GLP still relevant today, decades after its introduction? [U] / [CO1]

Even though Good Laboratory Practice (GLP) was introduced in the 1970s, it remains highly relevant and essential in modern scientific and regulatory environments. Its principles of quality, integrity, and traceability are foundational to trustworthy laboratory research.

9. What is the importance of eyewash station in a laboratory [C] / [CO5]

An eyewash station is a critical safety device in any laboratory where chemicals, biological materials, or particulates are handled. Its primary purpose is to protect the eyes from injury in case of accidental exposure to harmful substances.

10. What is the difference between save and save As? [C] / [CO5]

Both Save and Save As are commands used to store a file, but they serve different purposes in how and where your work is saved.

11. Write a short essay on raw data? [A] / [CO5]

Raw data refers to the original, unprocessed information collected during an experiment, observation, or study. It is the first-hand record of all measurements, observations, and facts obtained directly from the source before any calculations, interpretations, or analysis are made. In laboratories and research environments, raw data forms the foundation of scientific integrity and Good Laboratory Practice (GLP)**.**

12. How does proper log book maintenance prevent scientific misconduct? [U] / [CO4]

A log book (or laboratory notebook) is a formal record of all laboratory activities**,** observations, experiments, and results. Proper maintenance of a log book is a critical safeguard against scientific misconduct, as it provides transparency, accountability, and traceability for all research work.

13. Write four benefits of performing regular internal audit [K] / [CO5]
Four Benefits of Performing Regular Internal Audits1. Ensures Compliance with Standards and Regulations2. Identifies Errors and Process Inefficiencies3. Enhances Data Integrity and Reliability 4. Promotes Continuous Improvement and Risk Management

14. Why is it crucial to keep lab instruments dry and free from contaminants? [K] / [CO3]
Keeping laboratory instruments dry and contaminant-free is essential for accuracy, safety, and longevity of both the equipment and experimental results.1. Ensures Accurate and Reliable Results:2. Prevents Chemical Reactions and Damage:3. Maintains Equipment Longevity4. Prevents Cross-Contamination [4x5 = 20]

Part C

Long Essay Questions Answer any 2 questions Each question carries 10 marks

15. Critically analyze the effectiveness of email compared to other communication tools in research. [K] / [CO5]

Email has long been a primary mode of communication in research, but its effectiveness must be weighed against other tools like instant messaging, video conferencing, collaborative platforms, and project management software.1. Advantages of Email in Research a) Formality and Record-Keeping b) Asynchronous Communication c) Attachments and Detailed Information**.**

16. Evaluate the role of laboratory classification (BSL-1 to BSL-4) in ensuring biosafety.

[K] / [CO4]
Laboratory classification is the systematic categorization of laboratories based on the type of work, risk level, and safety requirements. Proper classification plays a critical role in ensuring safety, efficiency, regulatory compliance, and quality of research**.**

17. Develop a set of guidelines for safe and ethical use of the internet in laboratory research [U] / [CO5] The guidelines for the safe and ethical use of the internet in laboratory research address issues of data protection, human subjects, intellectual property (IP), and online collaboration. Researchers must maintain data security, uphold confidentiality, ensure proper consent, and give credit where it's due, regardless of whether their research is conducted online or offline.

18. Explain the importance of result retention and archiving in GLP studies. How does proper documentation affect the reproducibility of results? [U] / [CO1]
In Good Laboratory Practice (GLP) studies, result retention and archiving are critical for ensuring data integrity, regulatory compliance, and public safety. Proper, meticulous documentation directly enables the reproducibility of scientific results by providing a complete and reliable record of the study.

[10x2 = 20]